

# SmartDrive: Semi-Autonomous Scaled Robotic Car

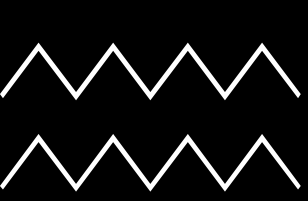
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Automotive Mechatronics I - Summer 2025

ECE4520/5520

Dr. Nashwan J. Sebi





# Approach – Technologies



Pixy2 Camera for color recognition,  
and distance calculations



LM393 module for RPM (Speed sensor)



FreeNove Four Wheel Drive Car Kit

- Frame
- Arduino
- Arduino Shield

# System Integration

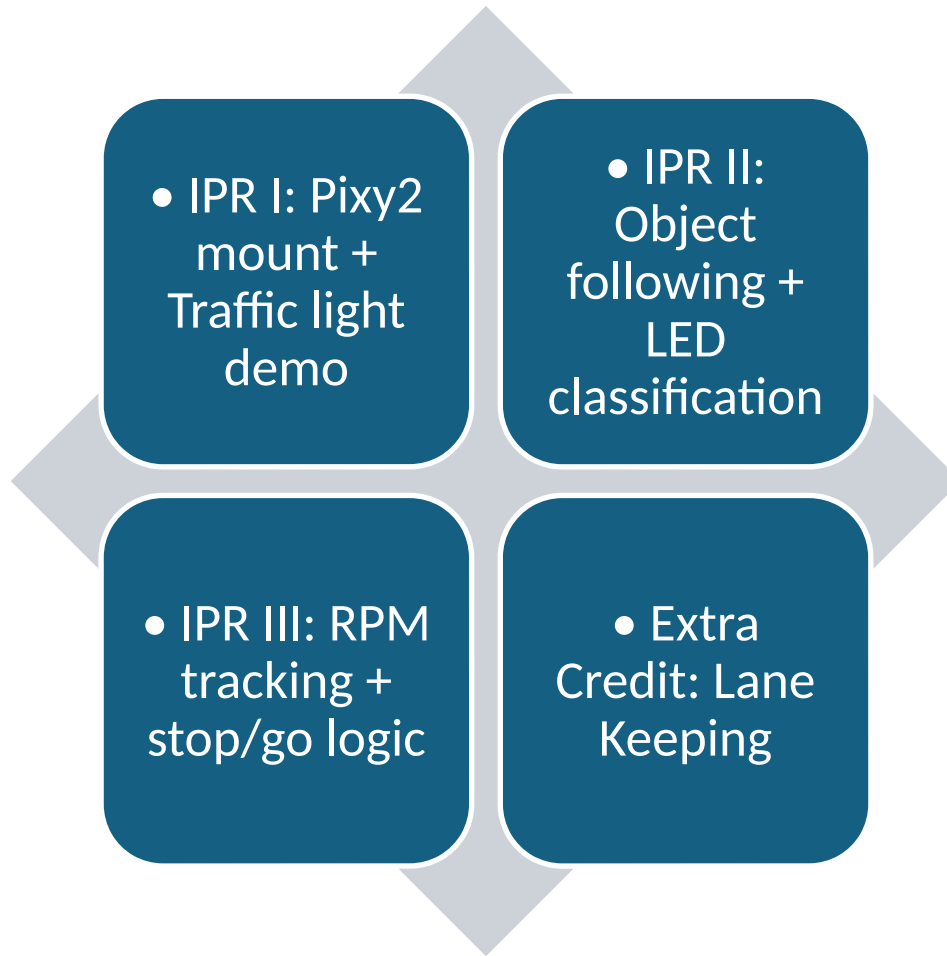
- Drive controlled via Arduino

- Pixy2 Software utilized

- PWM Speed control and LED brightness

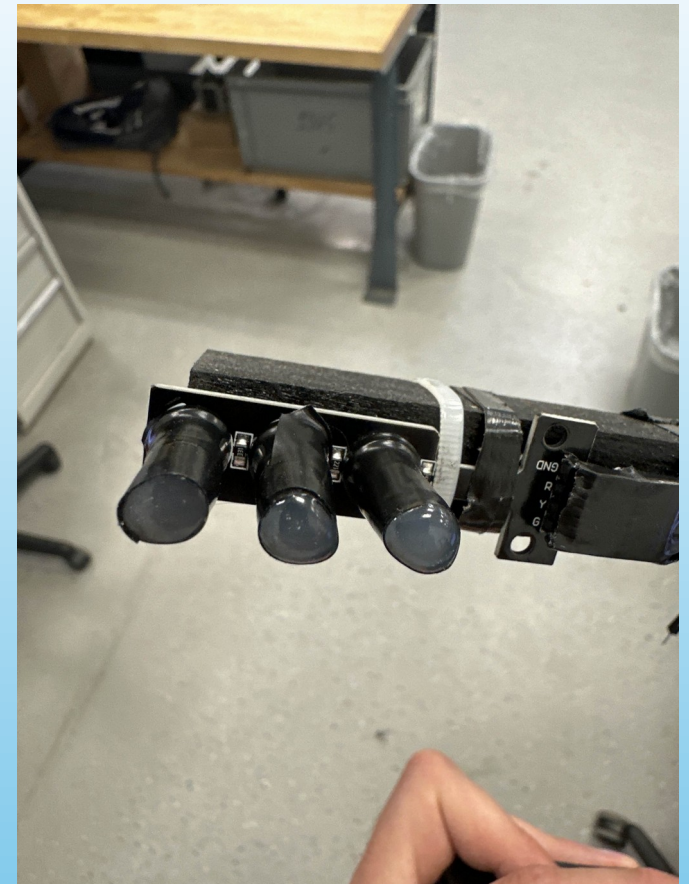
- Flow of data from vision → logic → actuation

# Tasks Overview



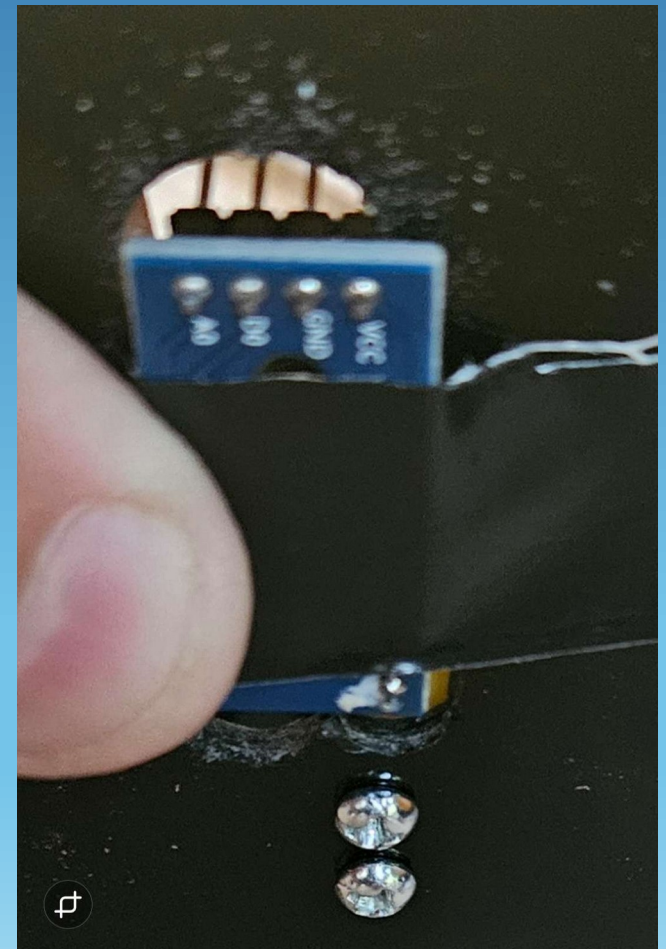


George  
finding out  
the use of  
the hot-glue  
gun resin





Eric installing  
the LM393  
Speed  
Measuring  
Module  
for RPM  
readings



# Control Logic Pseudocode

If sees RED → Slow → Stop at 2–6in

If GREEN → Maintain speed

If YELLOW + time < 8s → Stop  
-Else → Continue

If YELLOW + distance < 6" → Slow  
down

# Demo & IPR Videos

- Final Demo: [<https://youtu.be/T8GzPWGlayg>]
- IPR 1: [<https://youtube.com/shorts/pSG5HnREnpQ>]
- IPR 2: [<https://youtube.com/shorts/J1NlVZ9YNp0?feature=share>]
- IPR 3: [<https://youtu.be/CazVjC1R3RY>]
- Extra Credit: [<https://youtu.be/T8GzPWGlayg>]

# Milestones & Schedule

Week 9: Team  
kick-off (July  
5<sup>th</sup>)

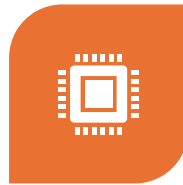
Week 11: IPR I  
Demo (July  
19<sup>th</sup>)

Week 13: IPR II  
Demo (Aug.  
2<sup>nd</sup>)

Week 14: IPR  
III Demo (Aug.  
9<sup>th</sup>)

Week 15:  
Extra Credit  
(Aug. 16<sup>th</sup>)

# Budget Summary



- PIXY2: ~ \$0  
A GIFT FROM PROF



- LM393 SPEED  
MODULE: ~ \$0  
A GIFT FROM PROF



- CHASSIS + MOTORS:  
\$58



- MISC: \$15



- TOTAL: ~ \$73